Question #1 of 17

Question ID: 1687656

Large trade sizes will have the largest effect on which of the following?

A) Delay costs.

B)

C) Opportunity costs.

Explanation

Large trade sizes, compared to the average daily volume, are likely to have market impact costs (a.k.a. price impact).

(Module 11.1, LOS 11.a)

Question #2 of 17

Question ID: 1687667

Which of the following is *most likely* to be used to reduce latency when trading?

A) Underclocked processors.

B) Simple operating systems.

C) Silicon processors.

Explanation

Electronic traders aim to reduce latency when trading by using the following:

- Overclocked processors that run faster than the designer intended
- Gallium arsenide processors rather than silicon processors
- Simple operating systems to avoid unnecessary programs slowing down response times

(Module 11.3, LOS 11.g)

Question #3 of 17

Question ID: 1687660

The quote for Stock Y is 40.21–40.27.

Which of the following statements is *most accurate*?

A) Buying at 40.21 means the effective spread equals the quoted spread. ✗

B) Selling at 40.20 means the effective spread is higher than the quoted spread. ✗

C) Buying at 40.26 means the effective spread is less than the quoted spread. ✓

Explanation

Buying at the ask price of 40.27 will result in the effective spread equaling the quoted spread. This is also the case when selling at the bid of 40.21.

Buying above the ask or selling below the bid results in the effective spread being greater than the quoted spread.

Buying below the ask or selling above the bid results in the effective spread being less than the quoted spread.

(Module 11.1, LOS 11.b)

Question #4 of 17

Question ID: 1687655

Which of the following is *most accurate* of the impact of electronic trading on transaction costs?

A) Lower transaction costs for large orders, but not for smaller orders. ✗

B) Lower transaction costs for large and small orders, and for institutional and retail investors. ✓

C) Lower transaction costs for institutional investors, but not for retail investors. ✗

Explanation

Electronic trading has reduced costs for both large and smaller trades, and for both retail and institutional investors. Large trades are more easily broken up into smaller trades for more efficient execution.

(Module 11.1, LOS 11.a)

Question #5 of 17

Question ID: 1687659

Consider a buy order for 12,000 shares of Stock P. A summary for the trade prices, sizes, prevailing bid, and offer prices are shown here:

Trade	Trade Price	Trade Size	Prevailing Bid	Prevailing Offer
1	15.22	4,000	15.18	15.22
2	15.29	8,000	15.19	15.23

Calculate the simple average effective spread for the overall share purchase.

A) 0.04.

B) 0.08.

C) 0.10.

Explanation

Effective spread = (trade price – midprice) × 2.

Trade 1:

= [15.22 – (15.18 + 15.22 / 2)] × 2

= (15.22 – 15.20) × 2

= 0.04.

If a purchase is made at the ask (or a sale is made at the bid), the effective spread will always be the same as the quoted spread. In this case, the quoted spread is as follows:

Quoted spread = 15.22 – 15.18

= 0.04.

Trade 2:

= [15.29 – (15.23 + 15.19 / 2)] × 2

= (15.29 – 15.21) × 2

= 0.16.

If a purchase is made at a higher price than the quoted ask (or as sale at less than the quoted bid price), the effective spread will be higher than the quoted spread.

Quoted spread = 15.23 – 15.19

= 0.04.

Simple average effective spread = 0.04 + 0.16 / 2

= 0.10.

(Module 11.1, LOS 11.b)

Question #6 of 17

Question ID: 1687664

Which type of trader attempts to exploit the option value of standing orders?

A) Electronic arbitrageurs. ✗

B) Electronic front runners. ✗

C) Electronic quote matchers. ✓

Explanation

Electronic quote matchers exploit the option values of orders waiting to be filled (a.k.a. standing orders). Quote matchers rely upon standing orders as a hedge against prices moving against them.

Electronic front runners are low-latency traders who use artificial intelligence methods to identify when other traders are trying to fill orders on the same side of the market. Front runners aim to anticipate order submissions, looking to identify predictable patterns to exploit.

Electronic arbitrageurs look across markets for arbitrage opportunities to buy undervalued and sell overvalued securities.

(Module 11.2, LOS 11.f)

Question #7 of 17

Question ID: 1687657

Which of the following is *most accurate* of market fragmentation?

A) It increases price disparity. ✓

B) It reduces price disparity. ✗

C) It has no effect on price disparity. ✗

Explanation

Market fragmentation is trading the same instrument in multiple venues. This increases the potential for price and liquidity disparities across venues, as the liquidity across each value varies.

(Module 11.1, LOS 11.b)

Question #8 of 17

Question ID: 1687670

A trader uses electronic arbitrage strategies. Which of the following is the *least* risky strategy?

A) Take liquidity on both sides. ✓

B) Offer liquidity on both sides. ✗

C) Offer liquidity on one side. ✗

Explanation

Taking liquidity on both sides involves using marketable orders to fill both legs. This is the costliest and the least risky of the three strategies.

Offering liquidity increases the risk but has the potential to reduce the cost.

(Module 11.3, LOS 11.g)

Question #9 of 17

Question ID: 1687663

Which trader is *most likely* to deploy natural language-processing techniques as the basis for trading decisions?

A) Electronic front runners. ✗

B) Electronic news traders. ✓

C) Electronic dealers. ✗

Explanation

Electronic news traders use natural language-processing techniques to automate the analysis of news data and to recognize trading signals. News traders must be fast—automating the analysis of the news speeds up this process.

Electronic dealers make a market by placing bids and offers, aiming to make the bid-ask spread.

Electronic front runners are low-latency traders who use artificial intelligence methods to identify when other traders are trying to fill orders on the same side of the market. Front runners aim to anticipate order submissions, looking to identify predictable patterns to exploit.

(Module 11.2, LOS 11.f)

Question #10 of 17

Question ID: 1687662

Which type of trader may typically trade in and out of a security more than 1,000 times during a single day in small sizes?

A) High-frequency trader. ✓

B) Low-latency trader. ✗

C) Electronic news trader. ✗

Explanation

High-frequency traders (HFTs) will typically trade in and out of a security within a minute—sometimes, within a few milliseconds. HFTs may typically continue this pattern in a security more than 1,000 times during a day in relatively small trade sizes.

Low-latency traders trade quickly on electronic news feeds and tend to hold their positions longer than high-frequency traders.

Electronic news traders need to react faster to new information to profit from their fast analysis.

(Module 11.2, LOS 11.f)

Question #11 of 17

Question ID: 1687669

Which of the following is created to reduce the latency of electronic trading?

A) Contingency table. ✓

B) Probability table. ✗

C) Profitability table. ✗

Explanation

Contingency tables contain prearranged defined actions in response to possible market movements. Identifying actions in advance removes the need for trader decisions and reduce delays.

(Module 11.3, LOS 11.g)

Question #12 of 17

Question ID: 1687666

Which of the following *best* describes the three needs for speed?

A) Faking, taking, and canceling. ✗

B) Canceling, making, and taking. ✓

C) Taking, making, and creating.

Explanation

There are three needs for speed:

1. Taking a trading opportunity before others do
2. Making bids early at attractive prices
3. Canceling orders that are no longer needed

(Module 11.3, LOS 11.g)

Question #13 of 17

Question ID: 1687671

Which of the following real-life events illustrates the systemic risks of a runaway algorithm?

A) May 6, 2010: the Flash Crash.

B) May 18, 2012: the Facebook IPO.

C) August 1, 2012: Knight Capital trading failure.

Explanation

A runaway algorithm results in high volumes of unintended orders resulting from programming errors. Knight Capital implemented a new trading system that produced a stream of unintended orders. Knight sent millions of orders to the markets over a 45-minute period when it planned to fill 212 orders. Knight lost $400 million due to the incident.

The 2010 Flash Crash was triggered by an overlarge order. The 2012 Facebook IPO illustrates the failure of a trading system due to two software processes locked into an infinite loop.

(Module 11.3, LOS 11.g)

Question #14 of 17

Question ID: 1687658

A portfolio manager reviews the following limit order book for shares of Instep, Inc. (INSP):

	Bid	Ask
Dealer X	88.87 for 500 shares	90.41 for 900 shares
Dealer Y	88.86 for 400 shares	90.45 for 700 shares
Dealer Z	88.81 for 600 shares	90.39 for 200 shares

Calculate the market spread.

A) 1.58.

B) 1.54.

C) 1.52.

Explanation

The market spread is the lowest ask – highest bid = 90.39 – 88.87 = 1.52.

The market spread is also known as the inside spread.

(Module 11.1, LOS 11.b)

Question #15 of 17

Question ID: 1687661

The trades for Stock Q during the trading day are as follows:

- Transaction 1: 3,000 shares sold at USD 20.20
- Transaction 2: 2,000 shares sold at USD 20.18
- Transaction 3: 8,000 shares sold at USD 20.12
- Transaction 4: 2,000 shares sold at USD 20.13

Trader X executed Transactions 2 and 3. **Calculate** the VWAP transaction cost.

A) USD 120.

B) USD 125.

C) USD 130.

Explanation

Stock Q VWAP for the trading day:

3,000 shares × 20.203,000 shares × 20.20	= 60,600
2,000 shares × 20.18	= 40,360
8,000 shares × 20.12	= 160,960
2,000 shares × 20.13	= 40,260
15,000	= 302,180
Weighted average price	= 302,180 / 15,000 = USD 20.145.

VWAP for Trader X:

2,000 shares × 20.18	= 40,360
8,000 shares × 20.12	= 160,960
10,000	= 201,320
Weighted average price	= 201,320 / 10,000 = USD 20.132.

VWAP transaction cost = shares sold × (Stock Q VWAP – Trader X VWAP)

= 10,000 × (20.145 – 20.132)

= USD 130.

(Module 11.1, LOS 11.b)

Question #16 of 17

Question ID: 1687668

Which type of computer code is *most likely* used to reduce latency for trading?

A) Python.

B) MS-DOS.

C) C++.

Explanation

Electronic traders optimize their computer code for speed and avoid slower high-level languages like C++ and Python. The original MS-DOS is simpler and faster.

(Module 11.3, LOS 11.g)

Question #17 of 17

Question ID: 1687665

Which of the following *best* describes the expectations of electronic news traders?

A) News traders expect successful trades most of the time.

B) News traders expect successful trades occasionally.

C) News traders expect successful trades half of the time.

Explanation

Electronic news traders trade with the expectation of only being right occasionally.

(Module 11.2, LOS 11.f)